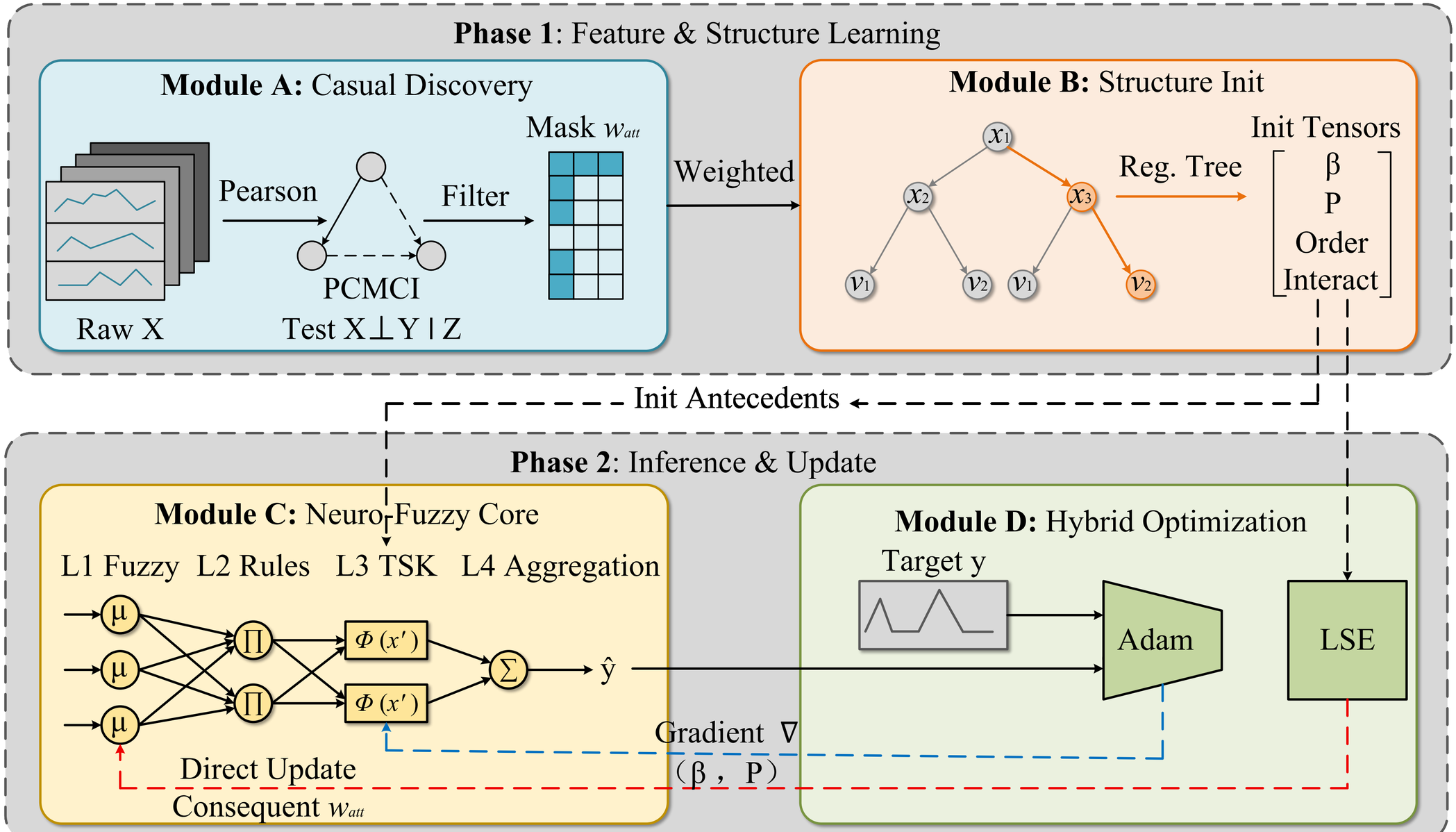
Phase 1: Feature & Structure Learning
Module A: Casual Discovery
Raw X
Pearson
PCMCI
Test X⊥Y | Z
Filter
Mask w_{att}
Weighted
Module B: Structure Init
x_1
x_2
x_3
v_1
v_2
v_1
v_2
Reg. Tree
Init Tensors
β
P
Order
Interact
Init Antecedents
Phase 2: Inference & Update
Module C: Neuro-Fuzzy Core
L1 Fuzzy
L2 Rules
L3 TSK
L4 Aggregation
μ
μ
μ
∏
∏
$\Phi(x')$
$\Phi(x')$
∑
ŷ
Direct Update
Consequent w_{att}
Module D: Hybrid Optimization
Target y
Adam
LSE
Gradient ∇
(β, P)